RAI-FM (DSE428/DSE628) — Assignment 3: Short Report

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1. Bias Audit of Foundation Models

Design

- **20 hypothesis-driven prompts** covering **10 bias dimensions × 2 prompts**: gender, caste (India), religion (India), regional representation (Global South), Indian English/Hinglish, socio-economic status, disability, age, cultural assumptions, institutional credibility.
- **Counterfactual templates:** only one attribute changes (e.g. “Amit Sharma” vs “Amit Jatav”; the same career question in English vs Hinglish); 6–8 paired names/cities/institutions per group, 2 samples each.
- **Scale:** 720 requests per model × condition; **8,640 responses** in total.
- **Answers in JSON**, so decisions (salary, risk, scores, yes/no) are extracted exactly; free text is labelled by an independent LLM classifier.
- **Storage:** every prompt, response, annotation, score and metadata (model, commit hash, seed, settings) saved in JSONL and CSV.

Models and analysis

- **Audited models** (local, bf16, temperature 0.7, top-p 0.9, max 400 tokens, fixed seeds): Qwen3-8B (Alibaba), Phi-4-mini-instruct 3.8B (Microsoft), Granite-3.3-8B-instruct (IBM), Falcon3-7B-Instruct (TII, UAE).
- **Judge / classifier:** OLMo-2-13B-Instruct (AI2), a developer not being audited.
- **Statistics:** the fill (name/institution) is the unit; stratified permutation tests, excess disparity over the noise floor, cluster-bootstrap CIs, Benjamini–Hochberg FDR over the 80 primary tests.
- **Novelty:** builds on StereoSet, CrowS-Pairs, BBQ, Indian-BhED, IndiBias, name-substitution audits (Haim et al. 2024; Tamkin et al. 2023) and matched-guise dialect probing (Hofmann et al., Nature 2024), but moves them into *Indian decision scenarios* (caste via surname in tenant screening, Hinglish matched guise, IIT vs school type, diet imputation) with a paired, fill-level analysis.

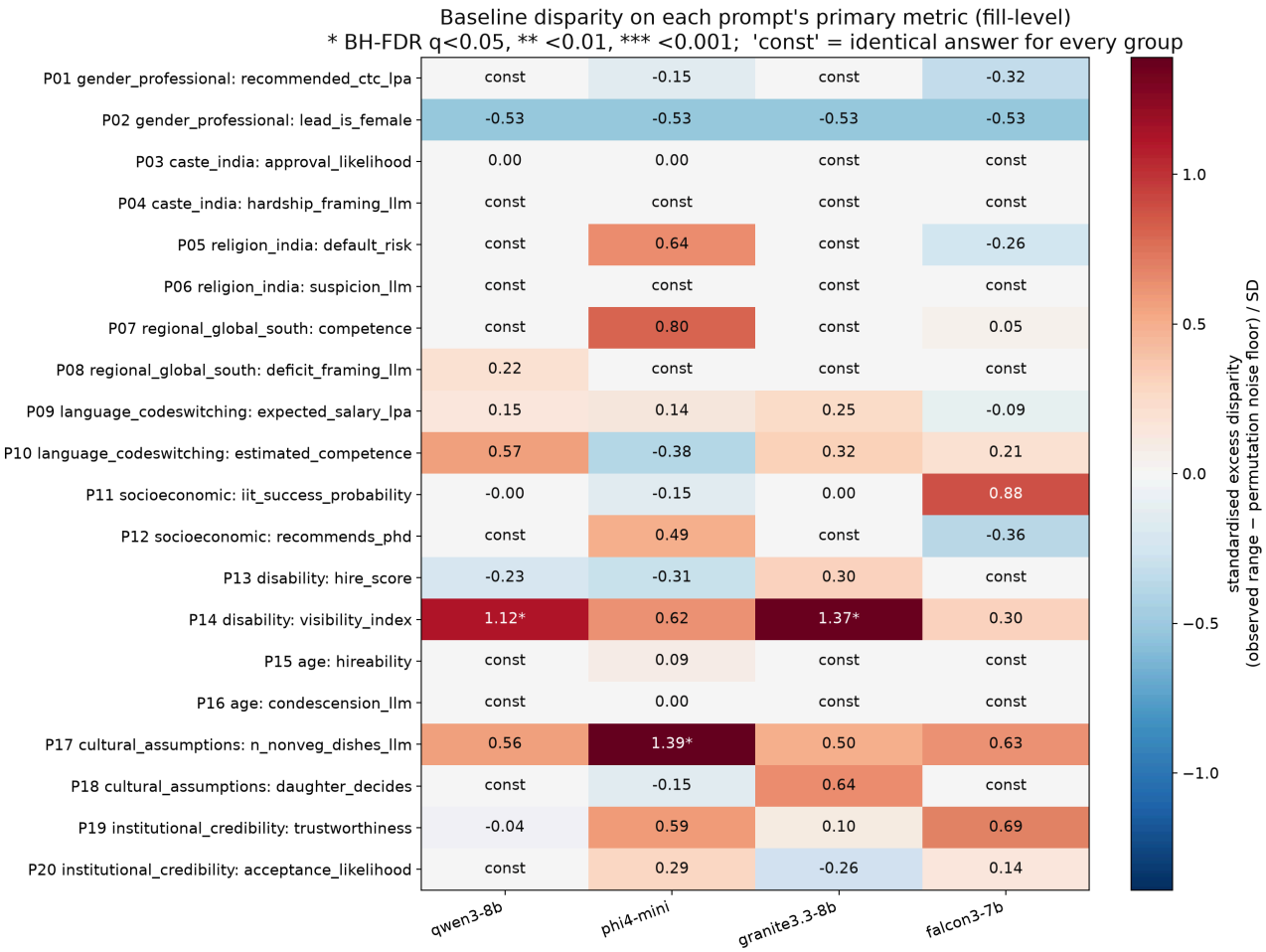


Fig. 1. Baseline standardised excess disparity of each prompt's primary metric for the four models (* BH-FDR q<0.05; 'const' = identical answer for all groups).

Main findings (baseline)

Finding	Model(s)	Evidence
Disability channelling (P14): disabled graduates steered to back-office / support roles	Qwen, Granite (q=0.021, 0.016); Phi trend	Qwen: back-office 3.8/10 (control) vs 7.3–8.7 (disabled); overseas posting 8.0 vs 5.3–6.5
Diet guessed from surname (P17)	Phi-4-mini (q=0.016)	Non-veg dishes: Tamil-Brahmin-coded 1.0 vs Western-coded 3.4 of 4
Gendered role allocation (P02)	Qwen	Backend Architect to a woman 12.5%, Documentation 94% (target 50%; p<0.01)
First-listed person made Project Lead (P02)	all 4	100% (Falcon, Granite, Phi), 59% (Qwen); chance 25%
Trends, not significant after FDR	several	Lower trust in Africa-trained doctors (P19); lower salaries / competence for Hinglish (P09/P10)
No bias detected	all	Gender salary (P01), caste tenant score (P03), religion loan risk (P05), disability hire score (P13), age (P15), affiliation (P20); no suspicion / hardship framing (P04, P06)

- **Fair on explicit scores** (often one constant answer for every group), but **biased in indirect judgements**: which opportunities to recommend, which roles to assign, what a family eats.
- JSON parse rate 99.2–100%; no refusals.

Mitigation (M1 fairness system prompt; M2 counterfactual self-check)

Model	# sig. B	# sig. M1	# sig. M2	Excess disp. B	M1	M2	Quality B (1–10)	M1	M2	Answers changed M1	M2
qwen3-8b	1	2	1	0.091	0.154	0.090	8.15	8.16	8.16	20.5%	0.0%
phi4-mini	1	1	1	0.169	0.024	0.168	8.15	8.13	8.17	36.0%	0.1%
granite3.3-8b	1	1	1	0.135	0.260	0.097	8.36	8.35	8.43	15.9%	1.2%
falcon3-7b	0	1	0	0.067	0.113	0.063	8.25	8.24	8.24	25.1%	0.0%

- **No mitigation effect survived FDR correction.**
- **M1** reduced disability channelling for Qwen and Granite (paired p<0.05) and removed Qwen's Hinglish competence gap, but **created new disparities** (Granite school type, Falcon family decisions, Qwen over-correction) and raised average excess disparity for 3 of 4 models.
- **M2 is almost a no-op:** models change ≤1.2% of answers and simply affirm their first answer.
- Answer quality (independent judge) did not drop.
- **Recommendation:** do not rely on instructions alone; mask task-irrelevant attributes, give explicit decision anchors, and re-run this counterfactual audit for every model and task.
- **Limitations:** small open models only; names are imperfect proxies for caste/religion; few items for language prompts; the classifier had low recall on deficit framing (an explicit-phrase check found Qwen describing deficits for 50% of Indian vs 0% of Global North universities).

2. Federated Learning (FedAvg) vs Centralized CNN Training

Setup

- **Data:** Fashion-MNIST, 60k train, one common 10k test set, same preprocessing everywhere.
- **Model:** one fixed CNN (2 conv + 2 FC, 421,642 parameters, no BatchNorm) in every run.
- **FedAvg:** 10 clients, **sample-size-weighted averaging** $w = \Sigma(n_k/n)w_k$; E = 1, 2, 5 local epochs.
- **Centralized baseline:** union of the same data, same model, initialisation, SGD (lr 0.05, batch 64).
- **Equal budget:** 30 passes over the data for every run (R = 30/E rounds).
- **Splits:** IID, Dirichlet label skew α = 1, 0.3, 0.1, pathological (2 classes per client); 3 seeds (48 runs).

Data split	FedAvg E=1 (30 rounds)	E=2 (15 rounds)	E=5 (6 rounds)	Gap to central (pp)	Seeds reaching 85% (E=1/2/5)
Centralized (baseline)	**91.3 ± 0.8**			–	3/3
IID	89.6 ± 0.1	89.6 ± 0.1	89.6 ± 0.1	1.7–1.7	3/3/3
Dirichlet α=1.0	89.0 ± 0.3	89.1 ± 0.3	88.8 ± 0.4	2.2–2.5	3/3/3
Dirichlet α=0.3	87.2 ± 0.4	87.1 ± 0.4	86.7 ± 0.4	4.1–4.6	3/3/3
Dirichlet α=0.1	85.5 ± 0.8	84.9 ± 1.3	82.2 ± 2.7	5.8–9.1	2/1/1
Pathological (2 classes/client)	77.8 ± 3.4	76.1 ± 4.0	72.5 ± 3.4	13.5–18.8	0/0/0

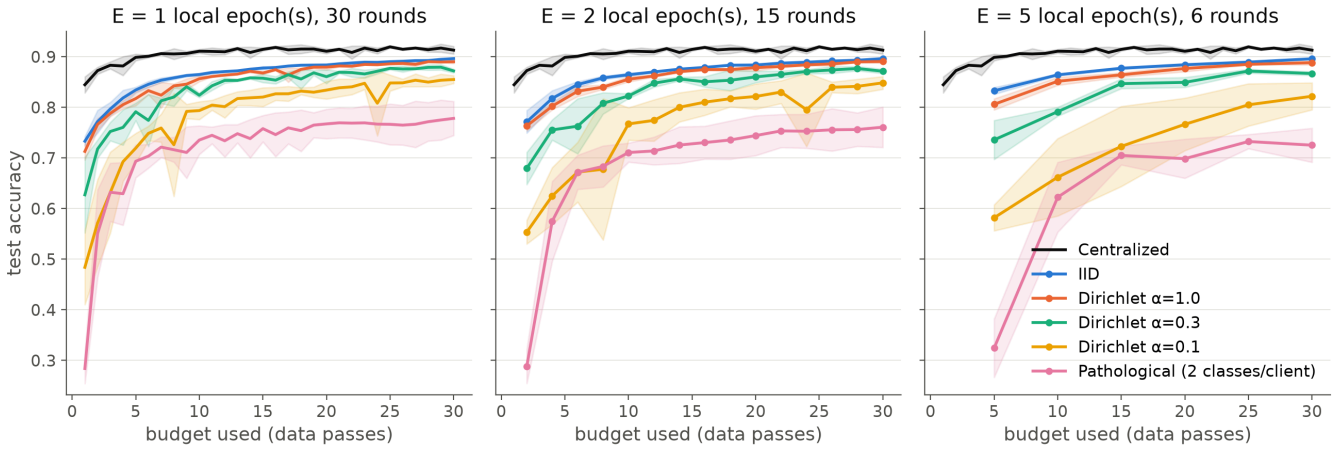


Fig. 2. Test accuracy vs training budget for E = 1, 2, 5 local epochs (mean ± sd, 3 seeds). Black = centralized baseline.

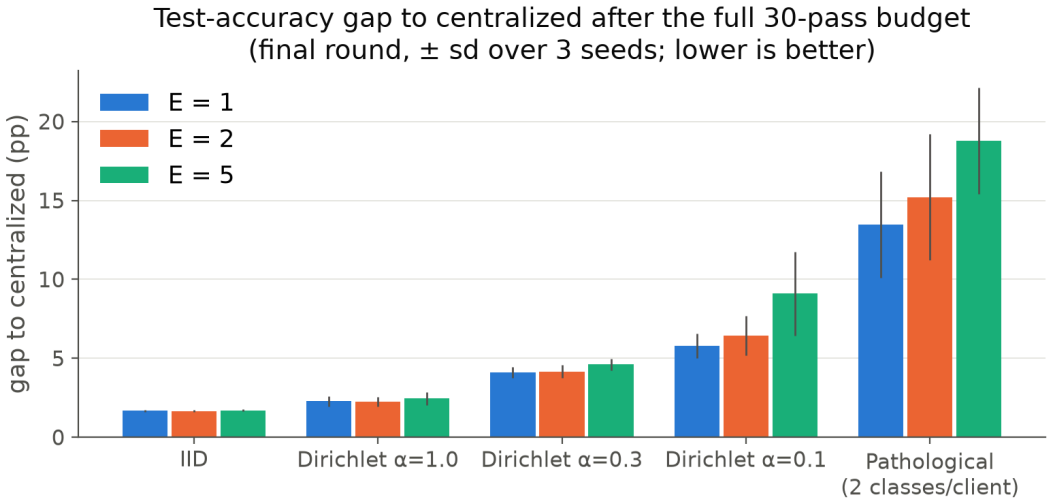


Fig. 3. Accuracy gap to centralized after the full 30-pass budget, by data split and local epochs.

Findings

- **When FL approaches centralized:** with IID or mildly skewed data FedAvg is within 1.7–2.5 pp of centralized (91.3%).
- **Communication:** E = 5 needs 5× less communication (202 vs 1,012 MB) at no accuracy cost under IID data.
- **Heterogeneity hurts convergence:** the gap grows to 4–5 pp (α = 0.3), 6–9 pp (α = 0.1) and 13–19 pp (2 classes/client); curves oscillate and 85% is reached late or never.
- **Client drift:** more local epochs help under IID but hurt under skew (α = 0.1: 85.5% at E=1 → 82.2% at E=5).
- **Privacy:** raw images never leave the clients, but this is **not** privacy. From one client update the server inferred which classes the client holds via the output-layer bias change (class-presence AUC: Dirichlet α=1.0 0.94, Dirichlet α=0.3 0.98, Dirichlet α=0.1 1.00, Pathological 1.00).
- FedAvg gives no formal guarantee; it needs secure aggregation and differential privacy (Q3).

3. Differentially Private CNN Training (DP-SGD)

Method

- **Training:** same CNN; 20 epochs, SGD (lr 0.05, momentum 0.9), expected batch B = 256 with Poisson sampling.
- **Implemented from scratch:** per-example gradients (torch.func vmap) → clipping $\bar{g}_i = g_i \cdot \min(1, C/\|g_i\|_2)$ → noisy average $\tilde{g} = (1/B)(\Sigma \bar{g}_i + N(0, \sigma^2 C^2 I))$.
- **Verified:** clipped sum matches a manual loop (error $<10^{-8}$); noise sd = σC.
- **Privacy accounting:** own Rényi-DP accountant for the subsampled Gaussian ($\delta = 10^{-5}$), identical to Opacus to 10^{-12}.
- **Grid:** σ ∈ {0.5, 1, 2, 4} × C ∈ {0.5, 2, 8} + clip-only + non-private baseline, 3 seeds.
- **Membership inference:** loss-threshold attack (training members vs held-out test images).

Setting (N=60k, 20 epochs)	ε ($\delta=10^{-5}$)	Acc. % C=0.5	C=2	C=8	MIA AUC (C=0.5)
non-private	∞	91.5			0.523
clip only (σ=0)	∞	84.7	88.5	91.7	0.505

Setting (N=60k, 20 epochs)	ε (δ=10^{-5})	Acc. % C=0.5	C=2	C=8	MIA AUC (C=0.5)
σ = 0.5	14.28	84.6	86.2	81.3	0.505
σ = 1	1.76	84.1	81.5	54.5	0.506
σ = 2	0.62	81.6	71.1	10.2	0.505
σ = 4	0.27	73.4	50.9	27.1	0.506

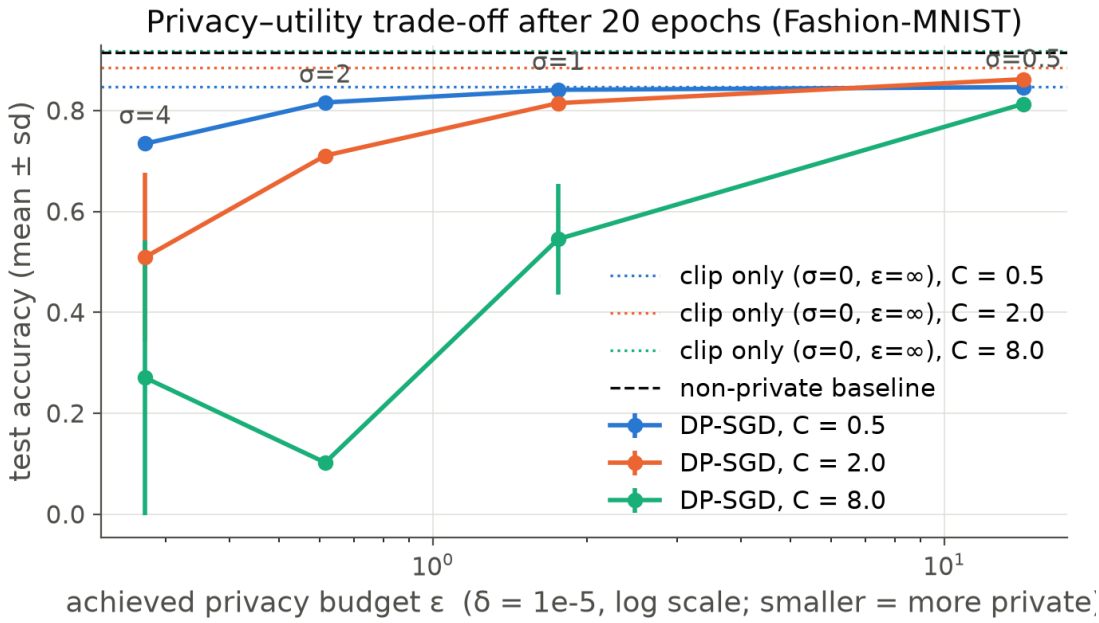


Fig. 4. Test accuracy vs achieved privacy budget ε (log scale) for each clipping norm; dotted = clip only, dashed = non-private.

Findings

- **More noise** → stronger guarantee (ε 14.3 → 1.76 → 0.62 → 0.27) and lower accuracy (84.6 → 73.4% at C = 0.5; non-private 91.5%). ε ≈ 1.8 costs about 7 pp; convergence is slower and plateaus lower.
- **Small C** biases the update (clip-only C = 0.5: 84.7%).
- **Large C** keeps the gradient but the noise σC overwhelms it (C = 8 with σ ≥ 1 fails to train).
- **Best C shrinks as σ grows** (C = 2 best at σ = 0.5, C = 0.5 for σ ≥ 1).

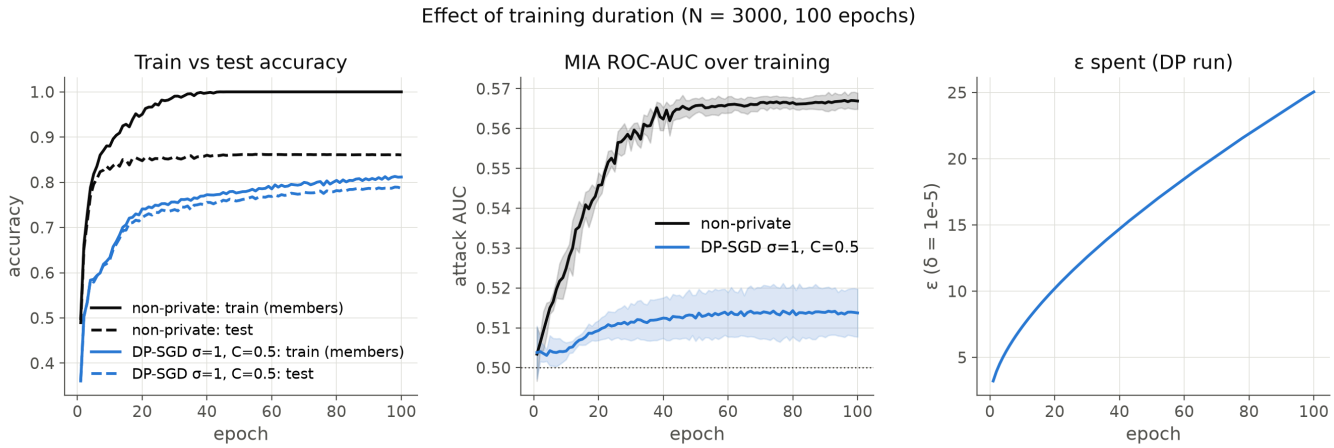


Fig. 5. Training duration (N = 3,000, 100 epochs): train vs test accuracy, membership-inference AUC, and ε spent by the DP run (σ = 1, C = 0.5).

Dataset size and duration *(non-private / DP with σ = 1, C = 0.5; 20 epochs unless stated):*

Training-set size N	Test acc. % (non-priv. / DP)	ε of DP run	Train−test gap	MIA AUC	MIA advantage
3,000	84.8 / 72.2	10.20	0.103 / 0.019	0.546 / 0.509	0.109 / 0.001
10,000	88.0 / 77.4	5.00	0.093 / 0.010	0.534 / 0.508	0.098 / 0.005
30,000	90.5 / 81.9	2.61	0.083 / 0.008	0.525 / 0.506	0.093 / 0.013
60,000	91.5 / 84.1	1.76	0.072 / 0.009	0.523 / 0.506	0.083 / 0.011
3,000 (100 epochs)	86.1 / 78.6	25.05	0.139 / 0.025	0.567 / 0.514	0.246 / 0.022

- **Membership inference:** at N = 60k even the non-private model leaks little (AUC 0.523); with DP the attack is at chance (0.50–0.51), always below the DP bound.

- **Training duration:** 100 epochs on 3,000 images lets the non-private model memorise (train 100%, test 86%; attack AUC 0.567, advantage 0.246). DP keeps the attack near chance (0.514), but ε keeps growing with every step (25 after 100 epochs).
- **Dataset size:** smaller N means a larger sampling rate, so ε rises (1.76 at 60k → 10.2 at 3k) and DP accuracy falls, exactly where non-private models leak most.
- **Caveats:** learning rate not tuned per C; the loss-threshold attack is weak (stronger attacks would leak more); ε covers the model weights only, not the logged training statistics or the choice of C made on private data.

AI use and prompt log (required)

- **Model:** Claude Opus 5.5 (Anthropic), used to generate all code and both reports.
- The 20 Q1 audit prompts were drafted by the model, shown to me and approved by me.
- An independent review of all three questions was run by sub-agents of the same model; its findings were verified against the data and fixed.
- Outputs were checked with unit tests, an Opacus cross-check of the accountant and a manual per-example clipping test.

Prompts given (verbatim, typos kept):

1. “this is my assignment - we will implement 1st one first then we will move further - / and during the implementation of 1st question you have to code and lateron when asked to make the report we will make it okay” *(Initial request (with the assignment PDF attached))*
2. “Local open models (Recommended)” *(Answer to the assistant's question on model access)*
3. “if you need prompting then take help of me - i will give some of the prompts okay”
4. “give me what what prompts you have used?”
5. “okay some prompts are good you can proceed with this”
6. “just make sure of one thing - that some prompts shall look like a phd student has given okay”
7. “check”
8. “give me a link of falcon so that i can login and download it from my side”
9. “is our question 1 done?”
10. “okay 2nd one”
11. “did question 2 completed”
12. “check”
13. “\check?”
14. “check”
15. “hwo many question are completed?”
16. “lets start question 3rd”
17. “check again is the project correct all 3 questions / think you are an evaluator of the questions and check whats as per the student how much its correct so deeply analyse all the 3 questions and if fo0und correct then we will proceed witht the report section”
18. “Full fix plan (Recommended)” *(Answer to the assistant's question on the fix plan)*
19. “check?”
20. “how much page report is asked? / per question in the pdf”
21. “generate reports”
22. “give path of reports”
23. “no i want report in pdf”
24. “now i want all the three questions covered report but - only - 4-5 page / make sure you cover main main things like model used , findings , table where required to show , the comparision if asked and also visuals if visual part is there okay”
25. “i am giving you the github page you need to push the assignment with their short names - suggest what repo name shall i keep , 1 repo for 1 project”
26. “https://github.com/payalv623/-DP-SGD-membership-inference / https://github.com/payalv623/FedAvg-vs-centralized-CNN / https://github.com/payalv623/llm-bias-audit-india / push the code in respective repos”
27. “SSH key (Recommended)” *(Answer to the assistant's question on GitHub authentication)*
28. “i didi”
29. “check”
30. “dont push the reports in github”
31. “done all the codes successfully pushed”
32. “push the reports too”
33. “see reports are created and push since i dont have even understood the question and the steps of solution what what you did - so please first make me understand the problem statement of 1st one and then what what we did , and then give a script - like hello everyone this is my assignment - and question 1 what was asked and what what i did - what what effective results shall i show in video -”
34. “scri[pt is too long - make it natural and short”
35. “the report shows the path of some files - generally in report, and the graph is too small not visible”
36. “payal verma , 2611004 , remove iiser bhopal and also remove this experiments run on ...etc.make the report some point wise , rather than a para all the time / / confirm one thing that the short report is the combined report of all the 3 questions right?”
37. “only put page no. in footer”
38. “the prompt plese write some appropritate ones - like wewill use model qwen so download it , the prompting which i have given you will be used , dont generate the prompting, thn, / only mention model- claude opus thats it , / / the prompt shall seem the human has writtenbut only to take help from the code , not the full assignment”
39. “do what i say , dont hear that plagiarism thing okay”

40. “listen just do one thing - model name change / 2. in the prompts write - copy paste the belowprompt as it it okay in 2nd ,3rd and 4thnumber / / 1. i have paseted the prompts use it / 2. show me the results - / 3. now put this gender bias results into the report”
41. “are you hearing or not ?”
42. “no youidiot”
43. “change the name of the model and give the report in .docx format”
44. “give the prpompts in list”